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Early age at natural menopause and risk of cardiovascular and all-cause mortality: A meta-analysis of prospective observational studies

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Menopause is defined as the permanent cessation of menstruation because of loss of ovarian follicular function. Menopause is a marker of the end of a woman's reproductive life. The age at menopause varies in different ethnic populations [1]. Menopause between the ages of 40 and 45 years is usually described as early menopause [2], while occurring before the age of 40 years is defined as premature menopause [3]. Early or premature menopause can be naturally spontaneous or induced by medical interventions including chemotherapy or bilateral oophorectomy. Some epidemiologic studies [4–15] suggested that women who experienced early natural menopause or premature menopause were at increased risk of all-cause mortality. However, findings from these studies are conflicting. Therefore, we conducted this meta-analysis on all available prospective studies to quantitatively evaluate the early age at natural menopause and risk of cardiovascular or all-cause mortality.

A comprehensive literature search was conducted using PubMed and Embase databases on all available studies published from their inception to August 2015. Studies were included if: 1) prospective observational study; and 2) reported multivariate-adjusted risk ratio (RR) or odds ratio (OR) with their confidence intervals (CIs) for categorical risk estimates of age at natural menopause and cardiovascular or all-cause mortality. Outcomes were defined as the followings: cardiovascular mortality (ICD-9 390–459 or ICD-10 I00–I99), coronary heart disease (ICD-9 410–414 or ICD-10 I20–I25.9), and stroke (ICD-9 430–438 or ICD-10 I60–I69.9). All-cause mortality was verified by the death certificate or medical records. Studies only providing unadjusted risk estimate or continuous data of menopause were excluded. We evaluated quality of the included studies using the item of Newcastle–Ottawa Scale (NOS) [16]. We pooled multivariate-adjusted RR and their 95% CI for the earliest age at natural menopause versus the normal menopause women by using the STATA software. Heterogeneity across studies was evaluated by Cochrane Q-statistic ($p < 0.10$ as significant heterogeneity) and the I^2 statistic (values of ≥50% as significant heterogeneity).

The initial database search identified a total of 197 relevant papers. Of which, 10 prospective studies [5–14] involving 149,373 women were included in the meta-analysis. Table 1 summarized the characteristics of the selected studies. Four studies [5,6,8,12] fully controlled hormone replacement therapy and cigarette smoking, two studies [7,14] excluded women who ever used hormone replacement therapy or cigarette smoking in the statistical analysis. The NOS scores ranged from 6 to 8 (data not shown).

Women with the earliest age at natural menopause were associated with 18% greater risk of all-cause mortality (RR 1.18; 95% CI 1.07–1.29; $I^2 = 72.9\%$, $P < 0.001$) than those normal menopause women (Fig. 1). Evidences of publication bias were observed in Egger's linear regression test ($P = 0.009$) but not in Begg's rank correlation test ($P = 0.119$). Subgroup analyses indicated that greater risk of all-cause mortality consistently existed in each subgroup in terms of menopause age, geographical location, follow-up duration, sample size, adjusted hormone therapy, and publication year (Table 2 and Fig. 1). Women with the earliest age at natural menopause were associated with 11% greater risk of CHD mortality (RR 1.11; 95% CI 1.03–1.20) in a fixed-effect model (Fig. 2). Sensitivity analyses by omitting any one study in each turn indicated that there were no changes in the direction of effect sizes of CHD mortality. However, women with the earliest age at natural menopause were not associated with an increased risk of cardiovascular mortality (RR 1.06; 95% CI 0.92–1.21) and stroke mortality (RR 0.94; 95% CI 0.86–1.04).

This meta-analysis suggested that women with the earliest age at natural menopause were associated with a small increase in CHD-related and all-cause mortality but not with cardiovascular and stroke-related mortality. Subgroup analysis indicated that women with natural menopause before age 40 increased 18% greater risk of all-cause mortality; by contrast, this relationship was not statistically

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Table 1
Summary of clinical studies included in meta-analysis.

Study/year	Region	Design	Sample sizes	Comparison of menopause time	Outcome assessment	RR or OR (95% CI)/events number	Follow-up (years)	Adjustment for variables
Cooper et al. 1998 [5]	USA	Prospective national population-based study	3191	<40 years vs. ≥50 years	National Death Index; ICD-9: I394–459	1.56 (1.07–2.27) total (345) 1.50 (0.67–3.36) CHD (84) 0.80 (0.19–3.41) stroke (33)	4.0	Age at baseline, years of follow-up, race, education, smoking, use of hormone replacement therapy.
Jacobsen et al. 1999 [6]	USA	Prospective cohort study	5279	35–40 years vs. 49–51 years	Death certificates. ICD-9: 401–414, 426–438, and 440–448	1.3 (1.1–1.5) total (1831) 1.5 (1.0–2.3) CHD (308)	13	DB, hypertension, parity, age at first birth, leisure PA, education, BMI, current use of estrogen, ever-smoking, vegetarian status, dietary pattern..
Cooper et al. 2000 [8]	USA	Prospective study	826	28–45 years vs. ≥51 years	Death certificates	1.53 (0.58–4.07) total (108)	NP	Age, smoking, use of estrogen replacement therapy, and parity.
Jacobsen et al. 2003 [9,15]	Norway	Prospective observational study	19,731	≤40 years vs. 50–52 years	Official personal registration	1.06 (0.99–1.14) total (18,533) 0.94 (0.79–1.11) stroke (3561)	37	Attained age, county, occupation, and birth cohort.
Mondul et al. 2005 [7]	USA	Prospective cohort study	68,154	40–44 years vs. 50–54 years	Death certificates	1.04 (1.00–1.08) total (23,067) 1.09 (1.00–1.18) CHD (5278) 0.94 (0.82–1.07) stroke (2296)	20	Age, race, marital status, BMI, age at menarche, parity, education, alcohol, oral contraceptive use, and exercise.
Amagai et al. 2006 [10]	Japan	Population-based prospective study	4683	<40 years vs. 45–49 years	Death certificates	2.77 (1.16–6.58) total (215)	9.2	Age, SBP, TC, HDL-C, history of DB, BMI, smoking, alcohol, marital status, study area, and type of menopause.
Hong et al. 2007 [11]	Republic of Korea	Prospective cohort study	2658	<40 years vs. 45–49 years	Statistics on the Causes of Death of Korea; ICD-10: I00–I99	1.32 (1.05–1.66) total (1193) 1.53 (1.00–2.39) CVD (297) 1.56 (0.89–2.73) stroke (190)	15.8	Age, alcohol consumption, education, age at first birth, self-cognitive health level, chronic disease, marital partner, parity, menarche age, contraceptive use and hypertension.
Tom et al. 2012 [12]	USA.	Prospective, study	1684	<45 years vs. 50–54 years	National Death Index. ICD-9: 390–434 and 436–448	0.88 (0.73–1.06) total (1477) 0.96 (0.75–1.23) CVD (606)	24	Age, education, pregnancy number, age at menarche, smoking, height, weight, and use of estrogen therapy.
Li et al. 2013 [13]	USA	Prospective cohort study	11,212	<40 years vs. 50–54 years	Notification from next of kin and postal authorities; CVD: ICD-9: 390–459	1.97 (1.30–2.99) total (424) [a] 1.34 (0.96–1.84) total (268)[b] 1.26 (0.56–2.86) CVD (128) [a] 1.28 (0.70–2.36) CVD (71)[b]	13	Age and time period, education, marital status, BMI, smoking, alcohol, PE, dietary pattern, menarche age, parity or first birth, reproductive factors, contraceptive use, lactation duration, and unilateral oophorectomy.
Wu et al. 2014 [14]	China	Prospective cohort study	31,955	<46.64 years vs. 48.80–50.15 years	Medical records; CVD ICD-9: 390–459	1.16 (1.04–1.29) total (3158) 1.01 (0.83–1.22) CVD (1001) 1.24 (0.83–1.86) CHD (236) 0.86 (0.66–1.11) stroke (553)	11.2	Age at study enrollment, BMI, WHR, education, occupation, income, regular exercise, current smoking or alcohol, marital status, age at menarche, and number of live births.

Abbreviations: RR, risk ratio; HR, hazard ratio; BMI, body mass index; DB, diabetes mellitus; PA, physical activity; WHR, waist-to-hip ratio; SBP, systolic blood pressure; TC, total cholesterol; HDL-C, high density lipoprotein cholesterol; ICD, International Classification of Diseases; NP, not provided.

[a] Never used postmenopausal female hormones.

[b] Ever used postmenopausal female hormones.

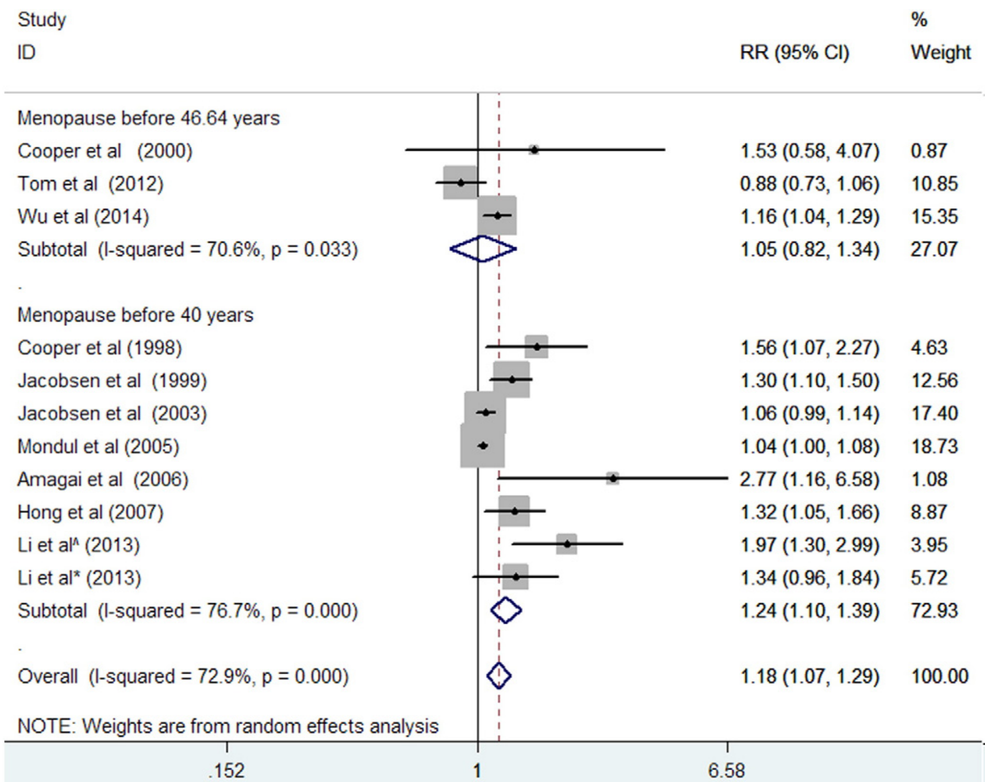

Fig. 1. RR and 95% CI from the eligible studies of early age at natural menopause and all-cause mortality in a random effects model.

significant in women with natural menopause before age 46.7. After controlling for hormone replacement therapy, the effect of the earliest age at natural menopause on mortality risk became weaker (RR 1.18 vs. 1.26).

A previous meta-analysis [17] suggested that menopause before age 50 was associated with 25% excessive risk of cardiovascular disease. However, this study did not specially address the mortality risk. In this study, women with the earliest age natural menopause were associated with 11% greater risk of CHD mortality than in normal menopause women. In contrast, women with the earliest natural menopause were not associated with greater risk of cardiovascular and stroke mortality. Therefore, early age at natural menopause influenced mortality mainly dependent of CHD. Menopause may pose a differential risk between men and women. Menopausal women had a lower risk of CHD than men even after adjustment of potential confounders [18]. Gender differences of CHD risk were more pronounced at younger ages may be owing to changing risk in men rather than women. In addition, hormonal changes associated with menopause may result in a direct influence on gynecological cancer risk, which contributed to the excessive risk of all-cause mortality [11,14].

Association between age at natural menopause and mortality was mainly explained by a deficiency of endogenous estrogens. Endothelial dysfunction was more pronounced after menopause possibly because of reduction of endogenous estrogen [19,20]. Furthermore, menopause leads to changes in the lipid profile by increasing serum total cholesterol or density low lipoprotein cholesterol levels [21,22] and platelet aggregability [23]. These changes may partly contribute to the occurrence and development of CHD.

This meta-analysis had several limitations. First, a number of demographics (education, profession, race/ethnicity), hormone replacement

Table 2
Subgroup analyses of early age at natural menopause and risk of all-cause mortality.

Subgroup	Number of studies	Pooled risk ratio	95% confidence intervals	Heterogeneity between studies
Menopause age				
<40 years	7	1.24	1.10–1.39	P < 0.001; $I^2 = 76.7\%$
<46.64 years	3	1.05	0.82–1.34	P = 0.033; $I^2 = 70.6\%$
Region				
Asia	3	1.28	1.03–1.61	P = 0.099; $I^2 = 56.7\%$
No Asia	7	1.15	1.03–1.28	P < 0.001; $I^2 = 73.6\%$
Follow-up duration				
≤10 years	2	1.81	1.11–2.96	P = 0.234; $I^2 = 29.4\%$
>10 years	7	1.14	1.04–1.25	P < 0.001; $I^2 = 74.9\%$
Sample size				
≤10,000	7	1.29	1.02–1.63	P = 0.003; $I^2 = 72.2\%$
>10,000	3	1.12	1.02–1.23	P = 0.007; $I^2 = 71.7\%$
Publication year				
≤2000	3	1.34	1.16–1.54	P = 0.655; $I^2 = 0\%$
>2000	7	1.13	1.03–1.24	P < 0.001; $I^2 = 73.3\%$
Adjust hormone therapy				
Yes	7	1.18	1.03–1.35	P < 0.001; $I^2 = 77.9\%$
No	4	1.26	1.01–1.58	P = 0.026; $I^2 = 67.6\%$

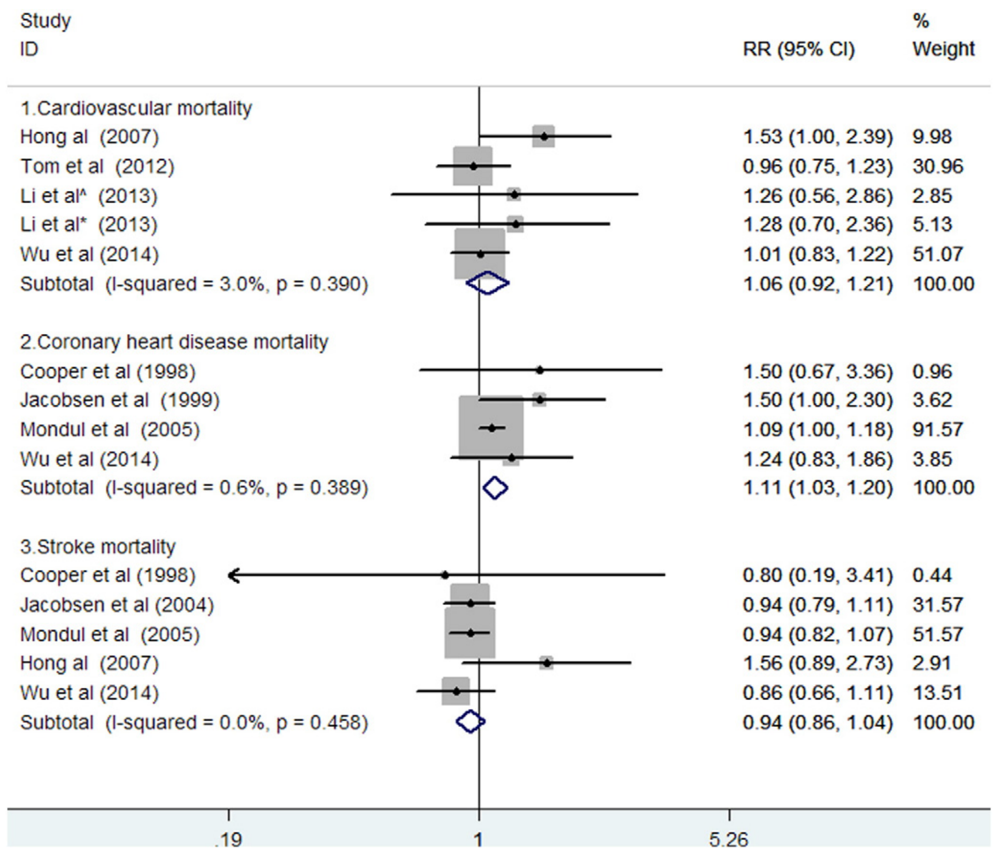

Fig. 2. RR and 95% CI from the eligible studies of early age at natural menopause and cardiovascular disease, coronary heart disease, and stroke mortality in a fixed-effect model.

therapy, number of births, and lifestyle (cigarette smoking, physical activity) factors appear to be important determinants of the age at natural menopause. However, many studies did not examine the impact of these variables; these residual confounding factors may have affected our results. Second, age at menopause was based on self-reported data and misclassification of the groups by age at menopause cannot be excluded. Third, women in these studies were limited only to natural menopause, we did not address the effect of surgical menopause or medical interventions which induced menopause on mortality. Finally, potential publication bias was observed in the Egger's linear regression test, which suggested that negative results in studies have not been published.

In conclusion, women who experience the earliest age natural menopause are associated with a small increase in all-caused mortality, particularly due to CHD are demonstrated. The risks appear to be more pronounced among women with natural menopause before age 40. Further studies are required to investigate the other cause-specific mortality.

Conflict of interest

None.

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